SEPARATEDNESS PROPERTIES DESCEND ALONG FINITE LOCALLY FREE GROUPOID QUOTIENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original proposition 07S6: sources/stacks-chow/tags.txt, source lines 9891–9891. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07S7: sources/stacks-chow/tags.txt, source lines 9892–9892. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02KQ: sources/stacks-chow/tags.txt, source lines 3381–3381. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02L6: sources/stacks-chow/tags.txt, source lines 3397–3397. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03GI: sources/stacks-chow/tags.txt, source lines 4529–4529. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04XV: sources/stacks-chow/tags.txt, source lines 6408–6408. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W6: sources/stacks-chow/tags.txt, source lines 2471–2471. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 03BS: sources/stacks-chow/tags.txt, source lines 4359–4359. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Notation and assumptions as in Proposition 07S6. Then*

(1) *if U is quasi-separated over S, then U/R is quasi-separated over S,*
(2) *if U is quasi-separated, then U/R is quasi-separated,*
(3) *if U is separated over S, then U/R is separated over S,*
(4) *if U is separated, then U/R is separated, and*
(5) *add more here.*

Similar results hold in the setting of Lemma 07S7.

Proof. Since M represents the quotient sheaf we have a cartesian diagram

$$\begin{array}{ccc} R & \xrightarrow[j]{} & U \times_S U \\ \downarrow & & \downarrow \\ M & \longrightarrow & M \times_S M \end{array}$$

of schemes. Since $U \times_S U \to M \times_S M$ is surjective finite locally free, to show that $M \to M \times_S M$ is quasi-compact, resp. a closed immersion, it suffices to show that $j : R \to U \times_S U$ is quasi-compact, resp. a closed immersion, see Descent, Lemmas 02KQ and 02L6. Since $j : R \to U \times_S U$ is a morphism over U and since R is finite over U, we see that j is quasi-compact as soon as the projection $U \times_S U \to U$ is quasi-separated (Schemes, Lemma 03GI). Since j is a monomorphism and locally of finite type, we see that j is a closed immersion as soon as it is proper (Étale Morphisms, Lemma 04XV) which will be the case as soon as the projection $U \times_S U \to U$ is separated (Morphisms, Lemma 01W6). This proves (1) and (3). To prove (2) and (4) we replace S by $\mathrm{Spec}(\mathbf{Z})$, see Definition 03BS. Since Lemma 07S7 is proved through an application of Proposition 07S6 the final statement is clear too. □